Beau Lonnquist

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EDUCATION

2025–2030 **University of California, San Francisco (UCSF)**, San Francisco, CA
Ph.D. in Biophysics (GPA: 4.00)

2021–2025 **University of Washington (UW)**, Seattle, WA
B.S. in Bioengineering & Data Science (GPA: 3.98)
Graduated *Summa Cum Laude* with Departmental Honors

RESEARCH FELLOWSHIPS

2025 NSF Graduate Research Fellowship (NSF GRFP)

2024 Washington Research Foundation Fellowship

2023 Levinson Emerging Scholar Fellowship

TECHNICAL EXPERIENCE

Research Experiences

2025–Present **Graduate Researcher**, Dept. of Bioengineering & Therapeutic Sciences, UCSF
Advisors: Brian Cheung & Tanja Kortemme

2022–2025 **Undergraduate Researcher**, Institute for Protein Design, UW
Advisors: David Baker & Cameron Glasscock (Rice University)

2021–2022 **Undergraduate Researcher**, Dept. of Mechanical Engineering, UW
Advisor: Katherine Steele

Industry Experiences

2024 **Cell Therapy Process Engineering Intern**, Genentech
Advisors: Rob Caren & Peter Ung

PUBLICATIONS

1. C. J. Glasscock, R. Pecoraro, R. McHugh, L. A. Doyle, W. Chen, O. Boivin, **B. Lonnquist**, E. Na, Y. Politanska, H. K. Haddox, D. Cox, C. Norn, B. Coventry, I. Goreshnik, D. Vafeados, G. R. Lee, R. Gordon, B. L. Stoddard, F. DiMaio, and D. Baker, "Computational design of sequence-specific DNA-binding proteins," *Nat Struct Mol Biol*, pp. 1-10, Sept. 2025, doi:10.1038/s41594-025-01669-4.

PRESENTATIONS

11. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Computational Design of *de novo* Transcription Factors for Targeted Genetic Repression," **Schrödinger's Molecules & Models Science Fair**, Virtual, 2025 (Keynote).

10. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Computational Design of DNA-Bending Tran-

scription Factors for Enhanced Genetic Control," **UW Bioengineering Capstone Showcase**, Seattle, WA, 2025 (Poster).

9. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Computational Design of DNA-Bending Transcription Factors for Enhanced Genetic Control," **UW Annual Undergraduate Research Symposium**, Seattle, WA, 2025 (Poster).

8. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Bending the Rules: de novo Transcription Factor Design for Targeted Gene Regulation," **Engineering Biology Research Consortium (EBRC) Annual Meeting**, Seattle, WA, 2025 (Poster).

7. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Computational Design of *de novo* Transcription Factors as Novel Gene Therapies," **Gulf Coast Undergraduate Research Symposium (GCURS) at Rice University**, Houston, TX, 2024 (Keynote).

6. **B. Lonnquist**, A. Gonzalez, and D. Slater, "Streamlining Stem Cell Therapy Manufacturing: Automated Approaches to Bioreactor-driven Stem Cell Expansion," **Genentech Annual Intern Poster Day**, Hillsboro, OR, 2024 (Poster).

5. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Synthetic Transcription Factors and Other Stories in Protein Science," **Genentech Hillsboro Innovative Therapies Lunch & Learn**, Hillsboro, OR, 2024 (Keynote).

4. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Computational Design of *de novo* DNA-binding Homodimers for Genetic Manipulation," **The Protein Society's 38th Annual Symposium**, Vancouver, BC, 2024 (Poster).

3. **B. Lonnquist**, C. J. Glasscock, and D. Baker, "Computational Design of *de novo* DNA-binding Homodimers for Genetic Manipulation," **UW Annual Undergraduate Research Symposium**, Seattle, WA, 2024 (Poster).

2. **B. Lonnquist**, C. J. Glasscock, R. Pecoraro, and D. Baker, "Unlocking Genetic Regulation: *de novo* DNA-binding Homodimers," **UW Summer Research Symposium**, Seattle, WA, 2023 (Poster).

1. **B. Lonnquist**, A. Lin, Z. Isley, S. Janakiraman, D. Nguyen, and K. Borgia, "Designing an Accessible Device for Card Games," **UW Center for Research and Education on Accessible Technology and Experiences (CREATE) Showcase**, Seattle, WA, 2022 (Keynote & Poster).

PEER REVIEW

1. **B. Lonnquist**, A. Fillion, J. Fraser, and B. Cheung, "PREreview of 'Zero-shot design of a de novo metalloenzyme,'" *Zenodo*, May 2026, doi: 10.5281/zenodo.20318681.

TEACHING & MENTORSHIP

2026–Present **PhD Curriculum Committee Representative**, Biophysics Program, UCSF

2020–Present **Mathematics Tutor**, Students Tutor Students

2024–2025 **Teaching Assistant–BIOEN 315**, Dept. of Bioengineering, UW

2023–2025 **Undergraduate Research Program Mentor**, Institute for Protein Design, UW

2022–2024 **Engineering Peer Educator**, College of Engineering, UW

2023 **Science & Engineering Instructor**, UW / Chehalis Foundation Summer STEM Camp

LEADERSHIP & COMMUNITY OUTREACH

2024–2025 **Mentorship Chair**, Biomedical Engineering Society, UW

2022–2024 **Vice President**, Triangle STEM Fraternity, UW

2022–2024 **Executive Board Member**, Husky Triathlon Club, UW

HONORS & AWARDS

Research Honors & Awards

2026 Hertz Fellowship Semifinalist (Interviewed) | Fannie & John Hertz Foundation

2025 Presentation Competition Winner | Schrödinger's Molecules & Models Science Fair

2025 Conference Travel Award (EBRC) | UW Office of Undergraduate Research

2024 Best Presentation in Biomedical Research | GCURS at Rice University

2024 Chemical and Biomolecular Engineering Travel Grant | GCURS at Rice University

2024 Poster Competition Winner | The Protein Society's 38th Annual Symposium

2024 Conference Travel Award (Protein Society) | UW Office of Undergraduate Research

Academic Honors & Scholarships

2025 Dean's Medal for Academic Excellence Finalist | UW College of Engineering

2024 Hoffman Endowed Scholarship | UW Dept. of Bioengineering

2024 Homi Kapadia Scholarship | Triangle Education Foundation

2023 Emerging Leader in Engineering | UW College of Engineering

2023 James Rust Scholarship | Triangle Education Foundation

2023 Stratos-Stephan Endowed Scholarship | UW Dept. of Bioengineering

2022 Edward J. Ammer Jr. Endowed Scholarship | UW College of Engineering

2022 Andy Grove Scholarship | Intel Corporation

2021 Purple and Gold Scholarship | UW College of Engineering

2021 Balanced Man Scholarship Semifinalist | Sigma Phi Epsilon at UW

2021 Cherie Pun Memorial Scholarship | Jacob Wismer Elementary School

Additional Honors & Awards

2025 Undergraduate Class Graduation Speaker | UW Dept. of Bioengineering

2024	Second Place, Jody Deering Nyquist Speech Contest | UW Dept. of Communication
2023	Outstanding Engineering Peer Educator of the Year | UW College of Engineering

Professional Affiliations

2025–Present	Tau Beta Pi Engineering Honor Society
2022–Present	Biomedical Engineering Society